



VIDYARTHI VIGYAN MANTHAN 2026-27

SYLLABUS (CLASS - X)

CLASS X

PHYSICS

- 1. Light:** Reflection of light by curved surfaces; Images formed by spherical mirrors, centre of curvature, principal axis, principal focus, focal length, mirror formula (Derivation not required), magnification. Refraction; Laws of refraction, refractive index. Refraction of light by spherical lens; Image formed by spherical lenses; Lens formula (Derivation not required); Magnification. Power of a lens. Functioning of a lens in the human eye, defects of vision and their corrections, applications of spherical mirrors and lenses. Refraction of light through a prism, dispersion of light, scattering of light, and applications in daily life (excluding the colour of the sun at sunrise and sunset).
- 2. Effects of Current:** Electric current, potential difference and electric current. Ohm's law; Resistance, Resistivity, Factors on which the resistance of a conductor depends. Series combination of resistors, parallel combination of resistors and its applications in daily life. The heating effect of electric current and its applications in daily life. Electric power, Interrelation between P, V, I and R. Magnetic effects of current: Magnetic field, field lines, field due to a current carrying conductor, field due to a current carrying coil or solenoid; Force on current carrying 9 conductors, Fleming's Left Hand Rule, Direct current. Alternating current: frequency of AC. Advantages of AC over DC. Domestic electric circuits.

CHEMISTRY

- Chemical Reactions and Equations.
- Acids, Bases and Salts.
- Carbon and its Compounds.
- Metals and Non- Metals.

BIOLOGY

- Life processes
- Control and coordination
- Reproduction
- Heredity
- Human eye
- Our environment

MATHEMATICS

- 1. Number Theory:** Divisibility theory in the Integers (The Division Algorithm, the Greatest Common Divisor, The Euclidean Algorithm, Fundamental Theorem of Arithmetic, Basic



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properties of congruence, sets theory, De Morgan's law, Ordered pair, Cartesian Product, Relations Linear congruences.

2. **Algebra:** Fundamental theorem of Arithmetic, Quadratic and Cubic Equation, Inequalities, Progressions (A.P, G.P) System of linear equations, Theory of equations, Binomial theorem and properties of binomial coefficients,
3. **Coordinate Geometry:** Area of Triangle, Quadrilateral, Straight Line, Distance, section formula,
4. **Geometry:** Triangle, Circles, Area of Circles, Tangent circle, Alternate segment theorem, Intersecting Secant theorem, Tangent- secant theorem, Ptolemy theorem, Pythagoras theorem, Basic Proportional theorem
5. **Trigonometry:** Basic Trigonometry, Trigonometric Function – Sum and Difference, Multiple angles (A , $2A$, $3A$, $A/2$), Height and distance
6. **Mensuration:** Volume and Surface area of Cube, Cuboid, Cylinder, Cone, Sphere, Hemisphere, 3 D figure including Pyramid and Prism, Frustum of a cone
7. **Probability:** Independent and dependent events, Situational and real-life probability problems

LOGIC AND REASONING

Pattern Recognition: Number Series, Alphabet Series, Image Series, Odd One Out, Analogy, Figure Matrix & Matrix Reasoning, Pattern-based Puzzles, Coding-Decoding, Logical Sequence of Words, meaningful word formation.

Analytical Reasoning: Blood Relations, Direction Sense, Order & Ranking, Seating Arrangement, Grid Puzzles, Games & Tournaments, Scheduling Problems, Clocks & Calendar, Sequencing & Ordering Problems, Sudoku.

Visual Reasoning: Mirror Images, Water Images, Rotation, Paper Folding, Paper Cutting, Figure Counting, Embedded Figures, Shape Construction, Spatial Visualization, 3D visualization, Cubes, Dice & Cube Nets, Best Fit, Symmetry.

Mathematical Reasoning: Mathematical Operations, Number Puzzles, Cryptarithmic (Alphametics) puzzles, Binary Logic, Logical Inequalities, Logical Venn Diagrams, Word Problems.

Critical Thinking: Data Sufficiency, Decision Making, Logical Connectives, Syllogism, Input Output series, Critical Path Analysis, Scheduling Problems.



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Emerging Reasoning Skills: Constraint Satisfaction Problems, Follow the Algorithm, Algorithm Output Prediction, Scientific Reasoning, Logical Puzzle Sets, Optimization, Geometrical reasoning.

LIFE STORY OF INDIAN SCIENTIST(S)

The Trailblazing Journey of Rohini Godbole (All Chapters)

INDIAN CONTRIBUTIONS TO SCIENCE

Indian Contributions to Science : Volume-III